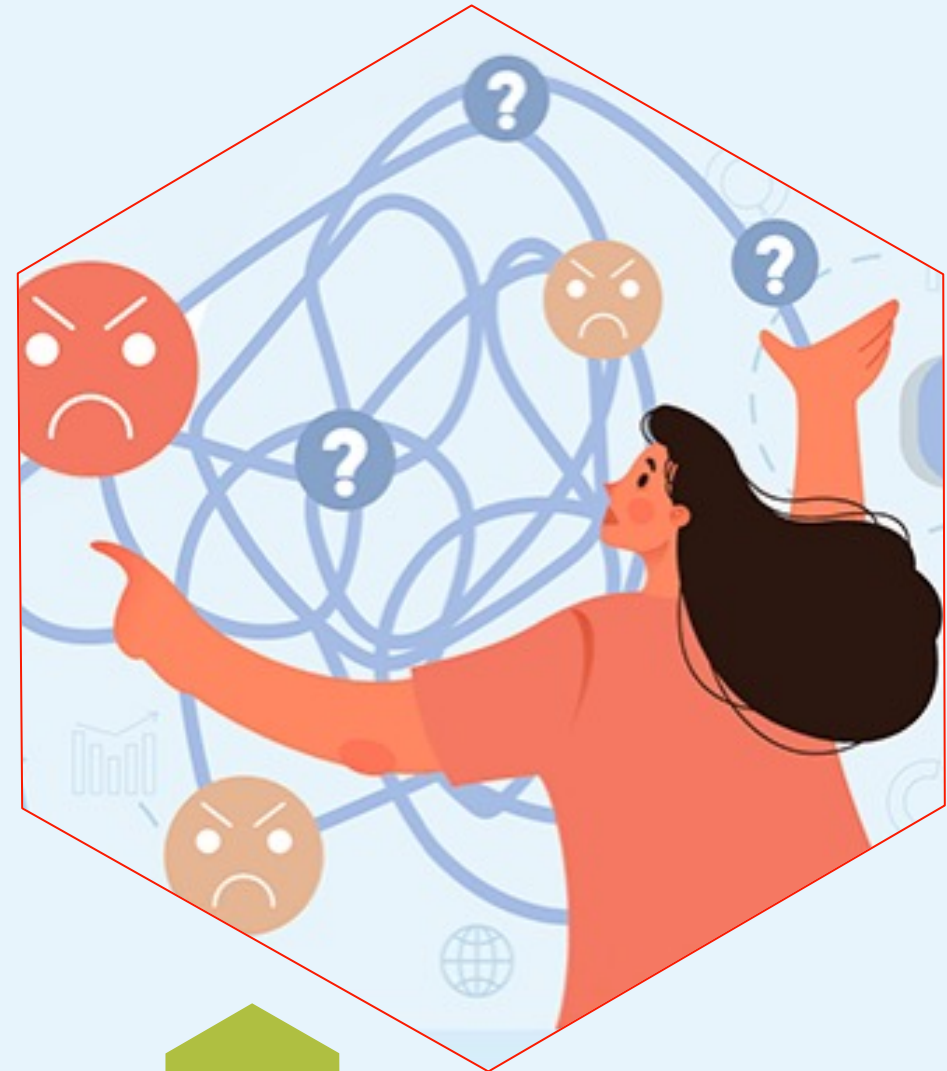


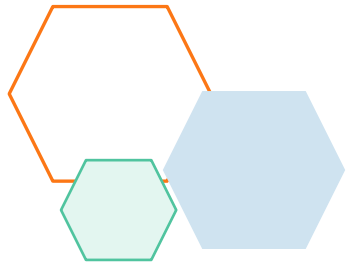
Lecture 4: Problem Framing & Modelling I

Dr. Mohammad Reza Bahrami

Faculty of Energy Engineering
Tashkent University of Technology
Introduction to Engineering
Fall 2026



Last Week, in One Breath



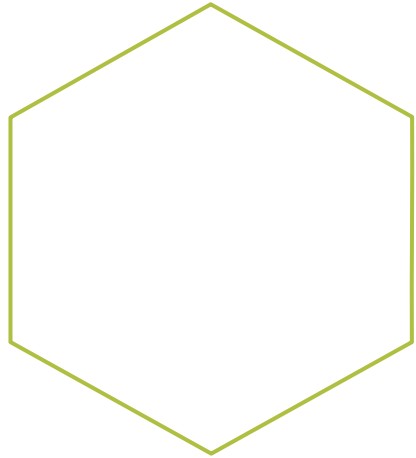
Systems — interacting elements and processes that turn inputs into outputs.

Process flow diagrams — show the steps inside the boundary and what flows between them.

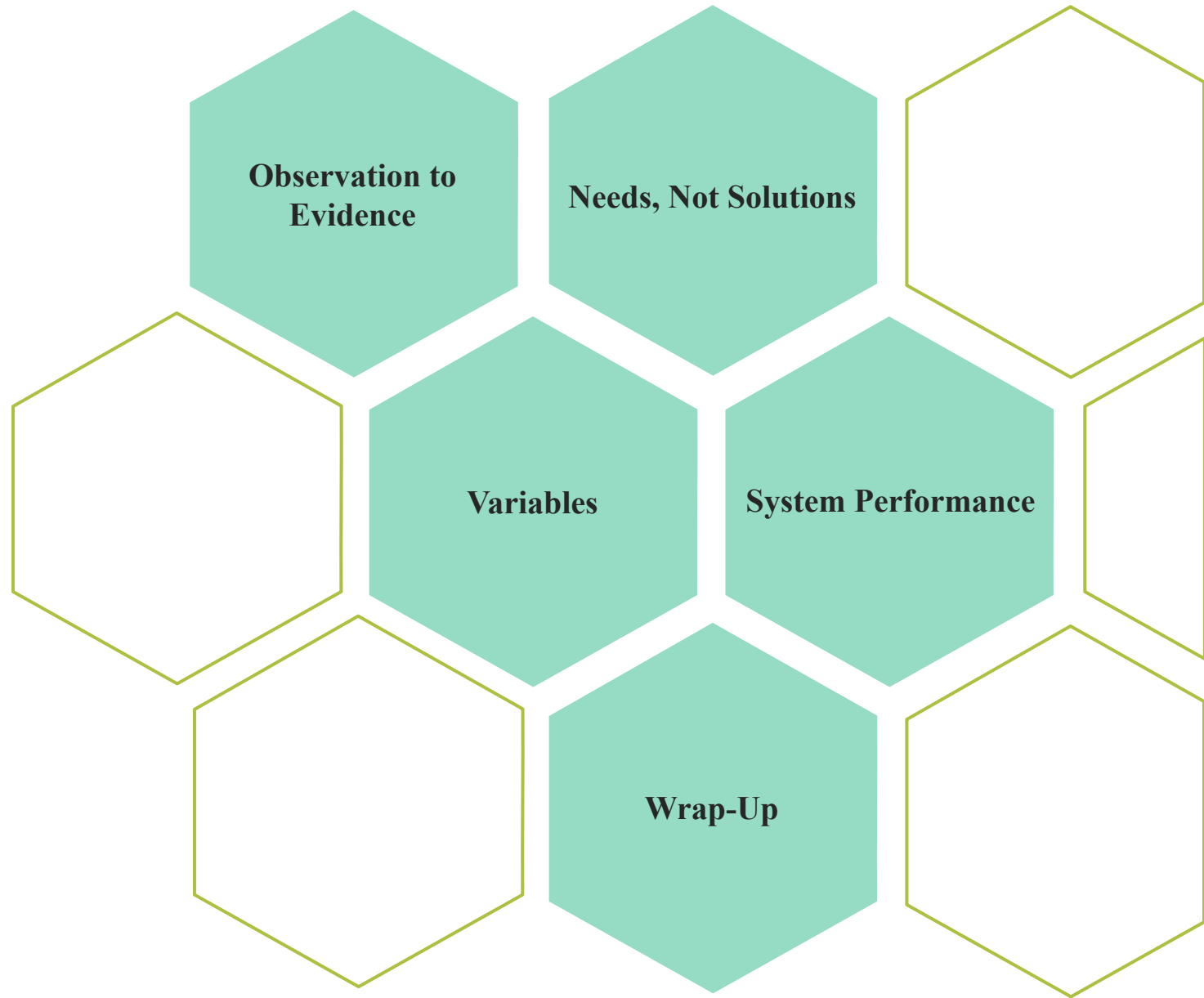
Perspectives — different stakeholders draw different diagrams of the same system.

Success — each stakeholder also defines “performing well” differently.

Today — how to turn those views into a clear problem and measurable variables.



Agenda



What Is the Problem?

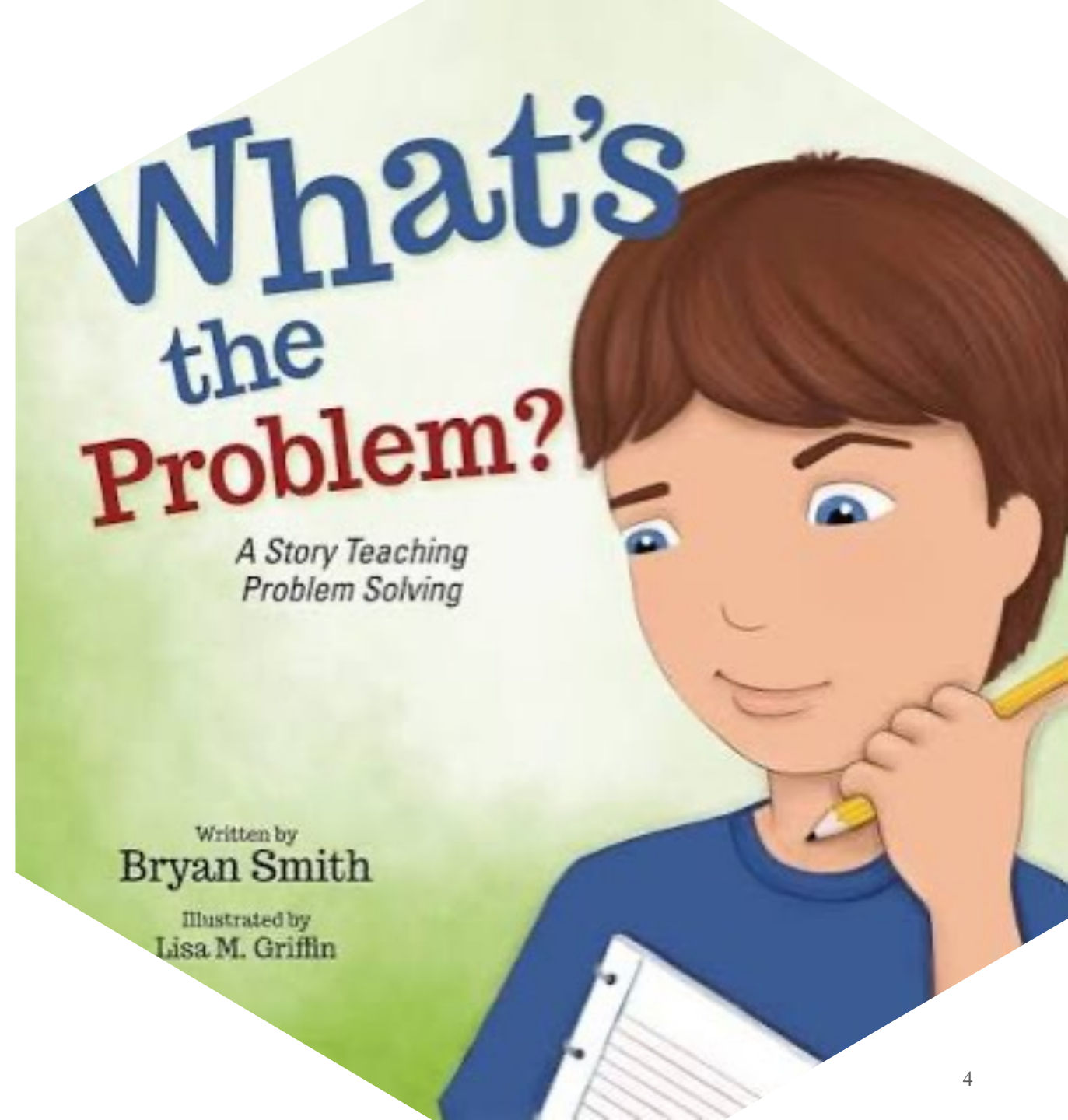
A fictitious client reports that people who work in a tall office building complain about long waits for the elevators.

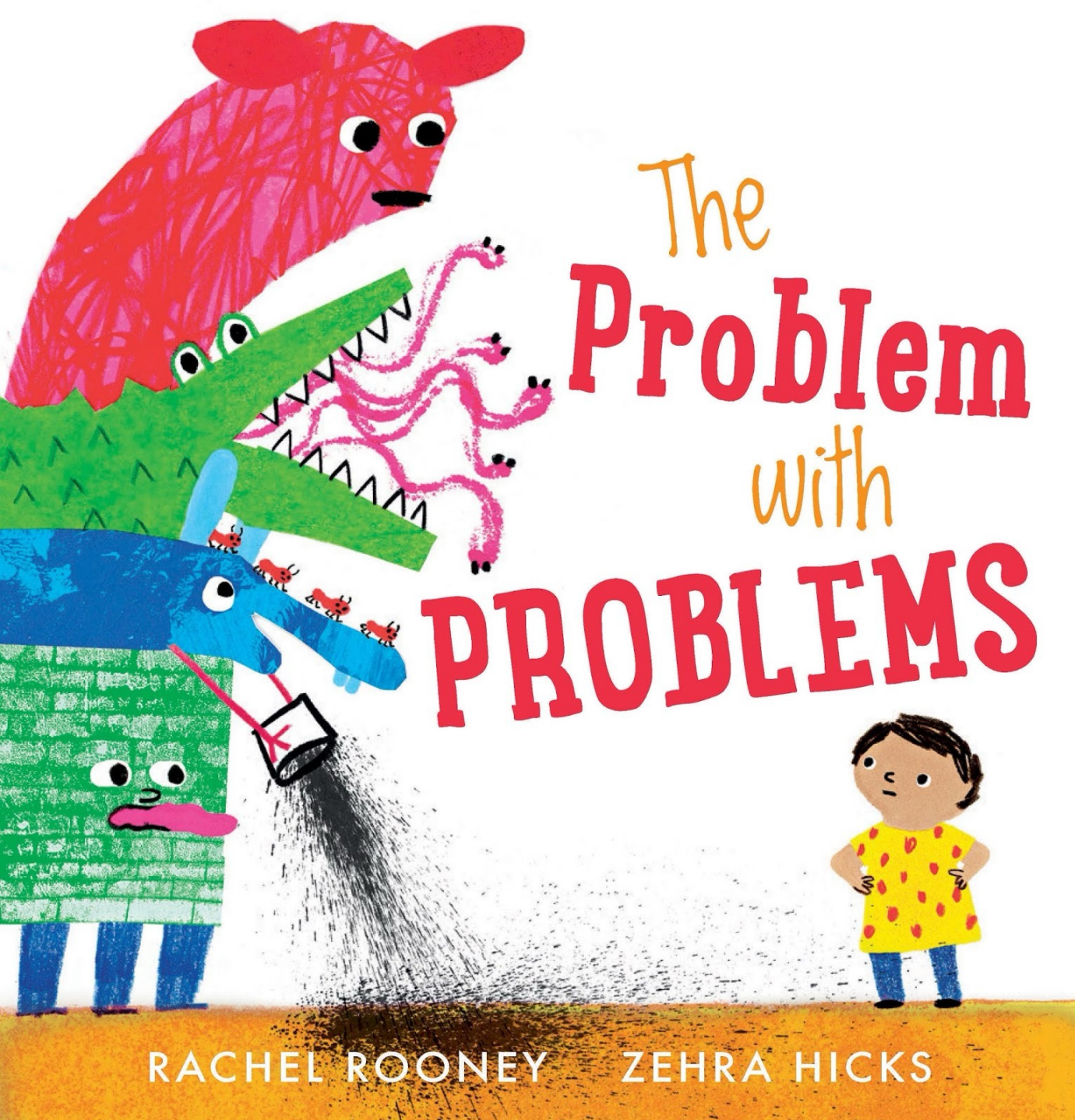
A: "Design a new elevator for the building."

B: "Increase employees' satisfaction with the elevators in the building."

Show of hands: A or B? Which one leaves more options open?

Source: adapted from Kosky et al., Exploring Engineering, Ch. 23 (Empire State Building example).





The problem with problems: They have a tendency to change our behaviors and make us do things that we instinctively know are wrong. It's almost as if there is some supernatural or paranormal power at work here.

To complicate matters, there are different types of problems and each requires a different tactic if we are to resolve them effectively.

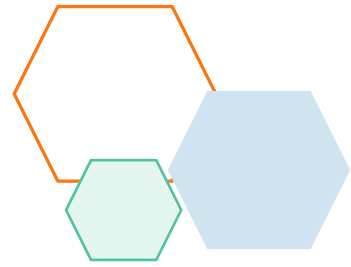
PART 1



From Observation to Evidence

What people tell us is raw material, not yet a problem.

What We Hear Is Messy



DINER

“The line is always too long.”

COOK

“We never get the delivery on time.”

MANAGER

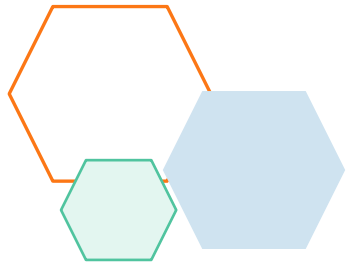
“Students waste too much food.”

NEIGHBOR

“The delivery trucks are so loud.”

Typical comments about the dining hall. Each one is a starting point, not a problem statement.

Before we define a problem, we ask: how do we know this is true?



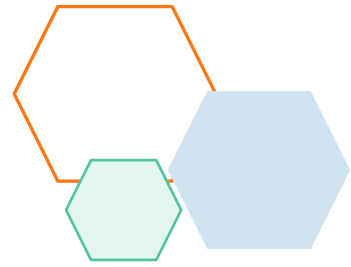
Four Kinds of Information

Kind	What it is	Example (dining hall)
Fact	Something observed or measured.	Lunch queues reach the door at 12:40 on Monday (counted).
Inference	A conclusion drawn from facts.	Queues are longest when two lectures end together.
Speculation	A guess without evidence yet.	The cashier may be the slowest step.
Opinion	A personal view or preference.	The hall should open another serving line.

All four are useful. Opinions and guesses show what people care about, and facts tell us what is happening.

Classification from Kosky et al., Ch. 23 (Clarifying the Problem); examples are ours.

Test What You Hear



1 Hear

Collect what people say,
including opinions and guesses.



2 Observe and measure

Watch the process. Count, time,
and ask stakeholders directly.



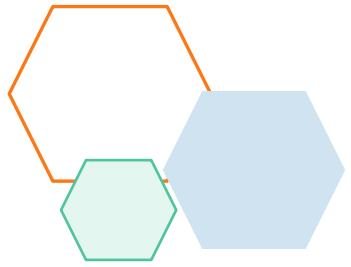
3 Decide

Keep what facts support.
Rethink what they contradict.

Example: many shoppers say a store needs more checkout lanes. Watching the line all day and talking to the manager may confirm that view, or show a different cause.

Your flow diagram from Lecture 3 tells you where to look: watch the steps, the waits, and the hand-offs.

Sort the Statements



In pairs, label each statement: fact, inference, speculation, or opinion.

A

“Lunch queues exceeded 15 people on 3 of 5 days this week.” (counted by a student)

B

“The dining hall is badly designed.”

C

“Queues are longer on days when the soup is served.”

D

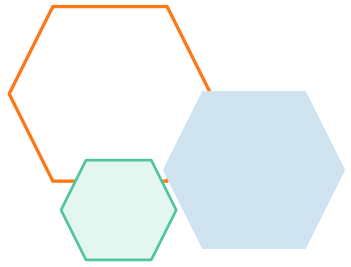
“I think the cashier is too slow.”

E

“The kitchen is probably running out of food early.”

F

“Average waiting time was 9 minutes at 12:30 on Tuesday.” (timed with a phone)



Sort the Statements: Answer Key

	Kind	Why
A	Fact	Counted, with a stated period.
B	Opinion	A judgement with no evidence.
C	Inference	A pattern read from observations.
D	Opinion	A personal view of the cashier.
E	Speculation	A guess about a cause, not yet checked.
F	Fact	Measured, with time and method.

Facts, inferences, speculation, and opinion each have a place, but only facts and good inferences can anchor a problem statement.

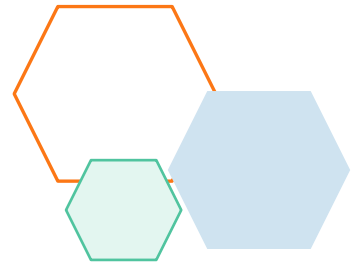
PART 2



Needs, Not Solutions

Writing a problem statement that keeps good answers open.

From Complaint to Problem Statement



COMPLAINT

What people say

“We wait too long for the elevators.”



NEED

What they require

Employees need to reach their floors without frustrating delays.

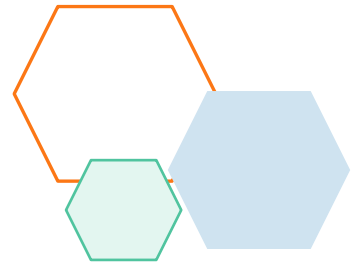


PROBLEM STATEMENT

What we will solve

Increase employees' satisfaction with the elevators.

A complaint describes a symptom. A need describes what the person requires. A problem statement turns the need into something we can work on.

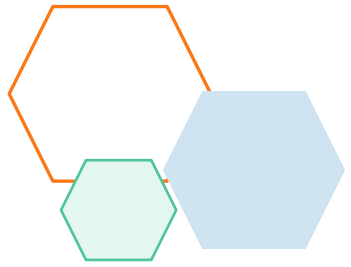


One Word Can Hide a Solution

Problem statement	Hidden solution	What it rules out
“Design a new elevator.”	A new elevator	Better scheduling, a mirror by the doors, free coffee on busy floors.
“Design a device to remove the blind spot.”	A device	Repositioning the mirrors, with no device at all.
“Design a blimp to fetch the ball.”	A blimp	An extendable arm that reaches the target.

The bias often sits in a single noun. Once the noun is in the statement, other answers are no longer considered.

Cases from Kosky et al., Ch. 23 (Empire State, blind spot, Blimp Wars).



A Problem Statement Template

COURSE TEMPLATE

[Stakeholder] needs **[outcome]** in **[system]**, judged by **[measure]**, within **[constraints]**.

Solution-free test

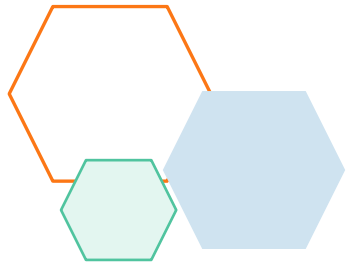
Does the statement name a device, app, vehicle, or building? If yes, rewrite it.

Example

Students need short, predictable waits for lunch in the dining hall, judged by waiting time at peak, within the current kitchen budget.

The template is a starting point. Adapt the wording, but keep the structure: who, what outcome, where, how judged, and what limits.

Rewrite the Problem



These statements each contain a solution. In pairs, find the hidden solution, then rewrite each as a need-based, solution-free statement.

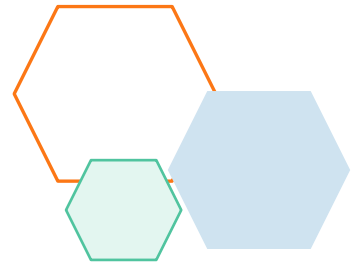
1 “Build a bigger dining hall.”

2 “Develop an app that shows bus arrival times.”

3 “Install more washing machines in the dorm laundry room.”

Use the template: [Stakeholder] needs [outcome] in [system], judged by [measure], within [constraints].

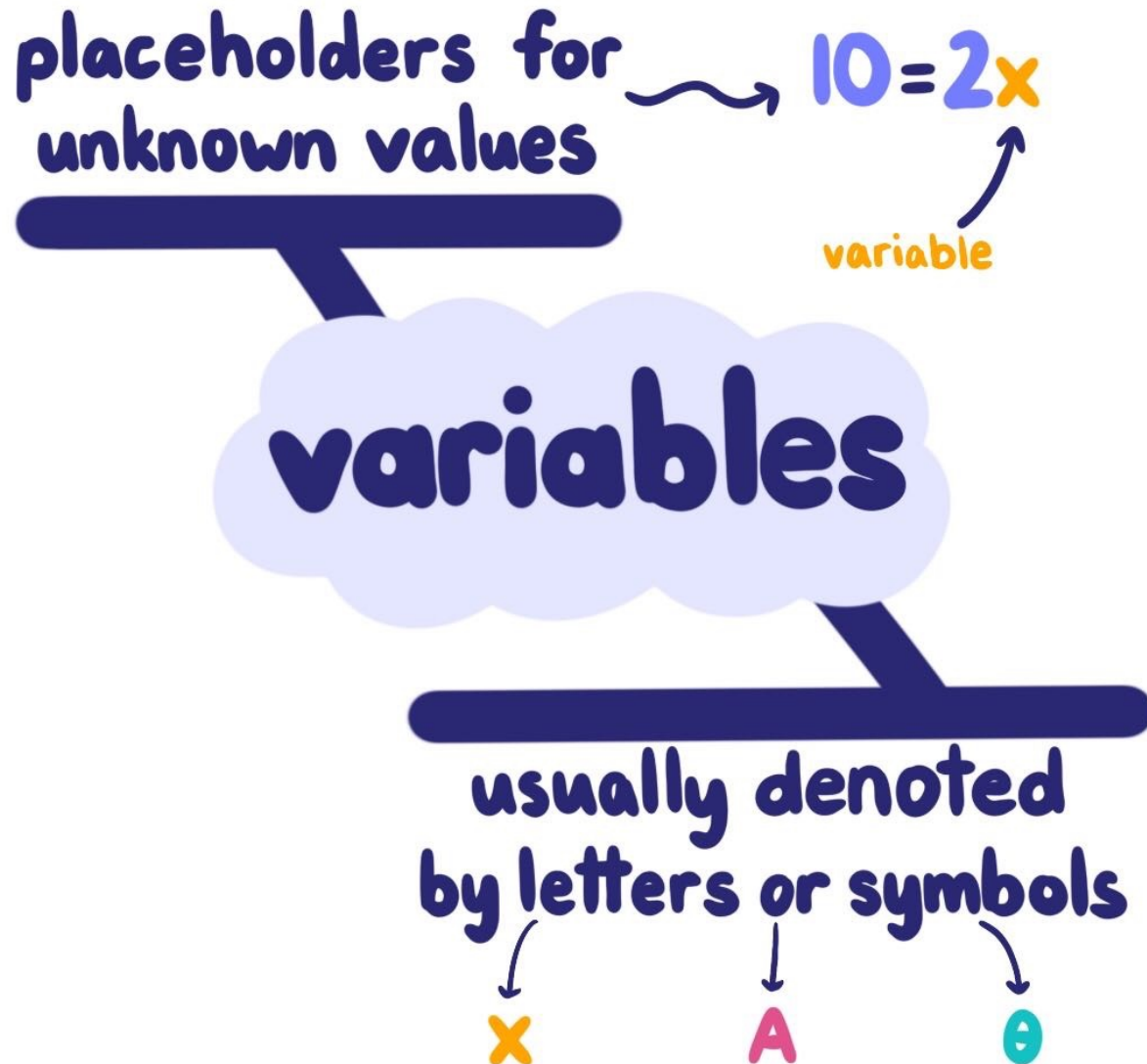
Rewrite the Problem: Sample Answers



Original	Hidden solution	One possible rewrite
“Build a bigger dining hall.”	A larger building	Students need to find a seat and eat lunch without long delays, judged by the time from entering to sitting down at peak.
“Develop an app that shows bus arrival times.”	An app	Riders need to know when the next bus will arrive, judged by the share of riders who can plan their wait.
“Install more washing machines.”	More machines	Residents need to wash clothes without long waits, judged by the time from arriving to starting a wash.

Many good rewrites exist. Check each for the solution-free test, not for matching this wording.

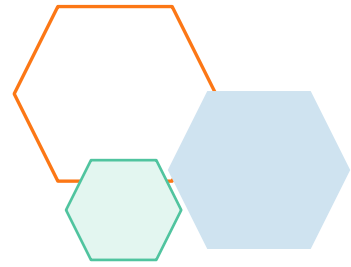
PART 3



Variables

Putting numbers on what matters, with the units to make them meaningful.

What Is an Engineering Variable?



A **variable** is a quantity we can measure, expressed as a number with **units**.

Time

measured with a clock, in seconds or minutes

Length

measured with a ruler, in meters

Temperature

measured with a thermometer, in °C or kelvin

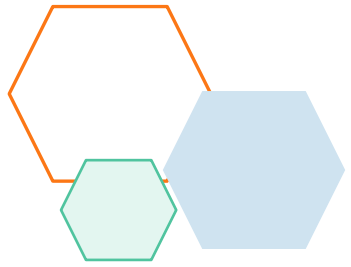
Count

how many students, meals, or buses

Speed, for example, combines a ruler measurement (distance) with a clock measurement (time).

Source: Kosky et al., Exploring Engineering, Ch. 2 (Engineering Variables).

A Number Without Units Is Incomplete



“Waiting time: 12”

12 seconds? 12 minutes? A student at the front of the queue and one at the back would read it very differently.

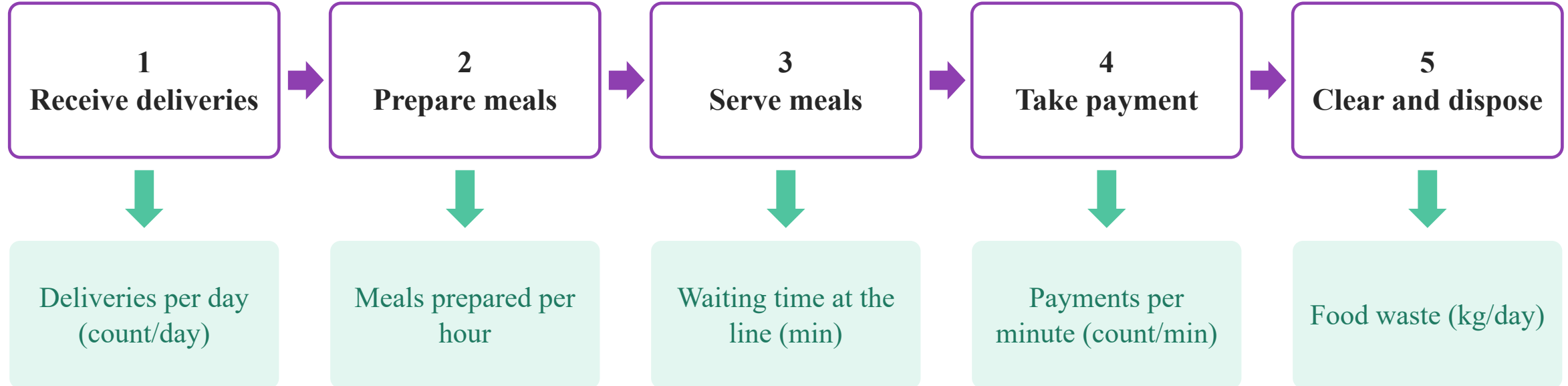
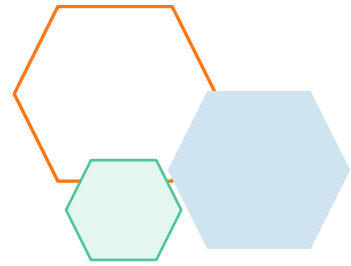
“Speed limit: 100”

100 km/h is about 62 mph. The same number means two different speeds.

In engineering a result has two parts: the value and its units. Both must be right.

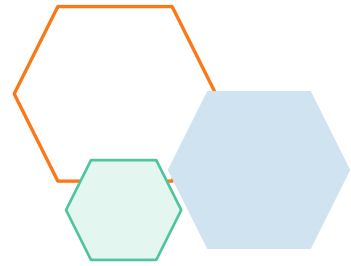
Idea from Kosky et al., Ch. 2 (Engineering Units of Measurement).

Variables Live on the Diagram



Every step and arrow in the Lecture 3 diagram suggests something we could count, time, or weigh.

Three Kinds of Variables



PERFORMANCE

What we judge

Average waiting time (min)
Meals served per hour
Food waste (kg/day)

DESIGN

What we can change

Number of serving lines
Staff on each shift
Menu size

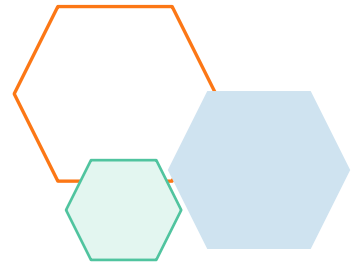
CONTEXT

What we cannot change

Opening hours
Health code limits
Building size

Performance variables tell us whether the system works. Design variables are the levers. Context variables are the limits.

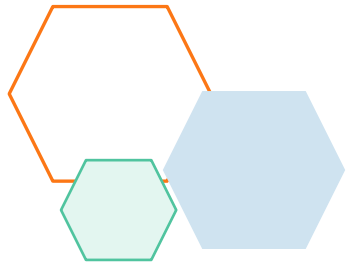
From Vague Words to Variables



Vague word	Variable we could measure	Unit
“Crowded”	People waiting in the queue	count, or people per m ²
“Slow”	Waiting time at the line	minutes
“Wasteful”	Food thrown away	kg per day
“Expensive”	Cost per meal	local currency per meal
“Unreliable”	Late deliveries per week	count per week

A useful variable answers “how much?” or “how many?”. If you cannot imagine measuring it, keep looking.

Find the Variables



- 1 Each table takes the campus bus route.
- 2 List 5 variables, with a unit for each: at least 2 performance, 1 design, and 1 context variable.
- 3 Say how each would be measured: what instrument, who, and when.

Hint: use your Lecture 3 diagrams, and think about the rider, the driver, and the university.

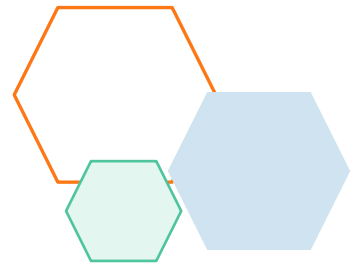
PART 4



Defining System Performance

Deciding what “working well” means, in numbers.

Stakeholders Suggest the Measures

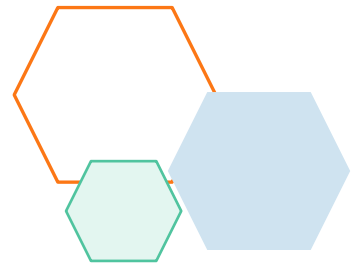


Stakeholder	Their idea of success	Candidate variable	Unit
Rider	Short, predictable waits	Waiting time at the stop	minutes
Driver / dispatcher	A schedule that can be kept	Trips completed on time	% of trips
Maintenance crew	Safe, reliable buses	Breakdowns per month	count/month
Administration	Good use of the budget	Cost per passenger	currency/rider
Neighbor	Less noise and traffic	Buses per hour on the street	buses/hour

These are the five perspectives from Lecture 3. Each one points to a different performance variable.

Illustrative candidates for discussion, not data.

Demands and Wishes



DEMAND (D)

Must be met

Without it the design is not acceptable.

Example: every bus passes a safety inspection before service.

WISH (W)

Nice to have

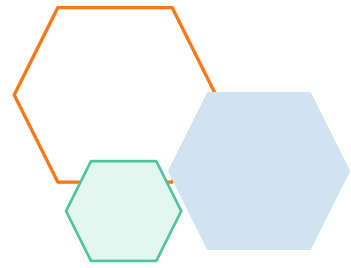
Desirable, but not crucial to success.

Example: real-time arrival information at every stop.

Use numbers wherever possible, even rough ones: “average wait at peak under 10 minutes”, not “short waits”. Numbers can be refined later. Illustrative values only.

Do not treat a wish as a demand. It can make the design more complicated than it needs to be.

Source: Kosky et al., Ch. 23 (List of Design Specifications).

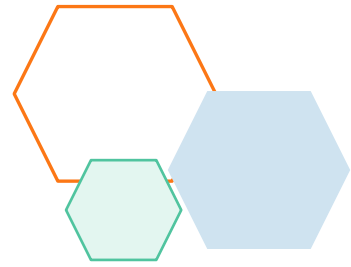


What Makes a Good Performance Measure?

- 1 Measurable.** You can count, time, or weigh it, with a stated method.
- 2 Relevant.** A real stakeholder cares about it.
- 3 Solution-free.** It does not assume how the problem will be solved.
- 4 Specific.** It has units, and a target, even a rough one.

Beware of measures that hide trouble: an average waiting time can look fine while some riders wait very long.

A Working Definition of System Performance



COURSE TEMPLATE

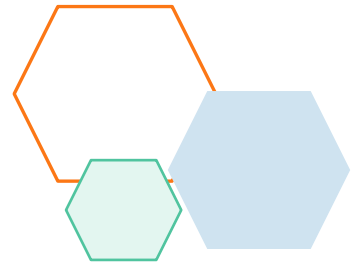
The performance of **[system]** is judged by **[variable 1]**, **[variable 2]**, and **[variable 3]**, each with units and a target, as seen by **[stakeholders]**.

Example: dining hall (illustrative)

Performance is judged by average waiting time at peak (min), meals served per hour (count/h), and food waste (kg/day), as seen by diners, kitchen staff, and administration.

“Working definition” means it can change as you learn more. Write it down so your team agrees on it.

From Problem Statement to Calculation



NEED

The variable we want to find

Average waiting time at peak

KNOW

What we already know

Arrival rate, number of serving lines

HOW

The method and assumptions

A simple flow or queue estimate

SOLVE

Do the calculation

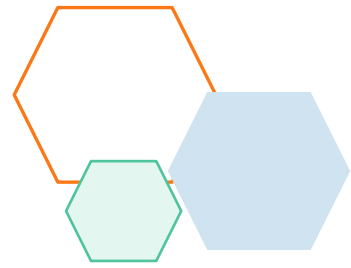
Compute, check units, interpret

Kosky et al. call this the Need-Know-How-Solve method. A clear problem statement and well-chosen variables give you the Need.

Next week: the How. Models are simplified representations, built on assumptions.

Method: Kosky et al., Ch. 2 (The Need-Know-How-Solve Method). Example values are illustrative.

Key Takeaways



Evidence before problem

Sort what you hear into fact, inference, speculation, and opinion.

Needs, not solutions

A problem statement should not name a device, app, or vehicle.

Variables carry units

A quantity we can measure, as a number with units.

Three kinds of variables

Performance (judge), design (change), context (limits).

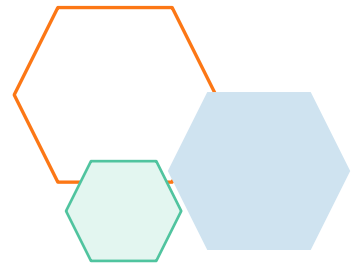
Stakeholders define success

Each perspective suggests different performance measures.

Write it down in numbers

Demands, wishes, and rough targets, refined as you learn.

Independent Study: Frame Your System's Problem



- 1 Write a one-paragraph problem statement for your system, in terms of needs and not solutions.
- 2 List 3 to 5 performance variables, each with a unit and how it would be measured.
- 3 Mark each as a demand or a wish, and give a rough numerical target.
- 4 Check your statement and variables against the solution-free test.

Nothing to hand in. This is working material for your team, and it prepares the next step: models and assumptions.



Thank You — Questions?

COMING UP NEXT

Lecture 5: Problem Framing & Modelling II

Engineering models as simplified representations, and how assumptions shape what a model can tell us.